

Section : Biology

Q.1 Which of the following are autosomal recessive trait/ disorder?

- (A) Thalassemia
- (B) Haemophilia
- (C) Sickle cell anaemia
- (D) Phenylketonuria

Choose the *correct* answer from the options given below:

- 1. (A), (B) and (D) only
- 2. (A), (C) and (D) only
- 3. (A), (B), (C) and (D)
- 4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.2 Which one of the following step is not involved in the genetic modification of an organism?

- 1. Identification of DNA with desirable genes
- 2. Cutting of DNA with DNA ligase
- 3. Introduction of the identified DNA into the host
- 4. Maintenance of introduced DNA in the host and transfer of the DNA to its progeny.

Options 1. 1

2. 2

3. 3

4. 4

Q.3 If white colour flower of *Antirrhinum sp.* successfully crossed with the pink colour flower, what would be the proportion of white and pink color flowers in the next generation?

- 1. 1:1
- 2. 3:2
- 3. 2:3
- 4. 2:1

Options 1. 1

2. 2

3. 3

4. 4

Q.4 Find the odd one out with reference to terrestrial ecosystem:

- 1. Forest
- 2. Grassland
- 3. Wetland
- 4. Desert

Options 1. 1

2. 2

3. 3

4. 4

Q.5 Which one of the following infections is completely curable?

- 1. Hepatitis -B
- 2. Gonorrhoea
- 3. Genital herpes
- 4. HIV

Options 1. 1

2. 2

3. 3

4. 4

Q.6 Which of the following are causes of biodiversity losses?

- (A) Habitat loss and fragmentation
- (B) Alien species invasions
- (C) Zero exploitation
- (D) Co-extinctions

Choose the *correct* answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (C) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Which one of the following diseases can be treated by human protein (α -1-antitrypsin) produced by transgenic animals?

1. Cystic fibrosis
2. Cancer
3. Emphysema
4. Haemophilia

Options 1. 1

2. 2
3. 3
4. 4

Q.8 Which of the following are not an Intra-Uterine device?

- (A) Lippes loop
- (B) Pills
- (C) LNG 20
- (D) Diaphragms

Choose the *correct* answer from the options given below:

1. (B) and (D) only
2. (A), (B) and (C) only
3. (A) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Identify the genetic disorder in which affected individual has a furrowed tongue, retarded mental development and broad palm?

1. Klinefelter's Syndrome
2. Down's Syndrome
3. Turner's Syndrome
4. Haemophilia

Options 1. 1

2. 2
3. 3
4. 4

Q.10 Match List-I with List-II

List-I

List-II

Male reproductive parts Functions

- | | |
|------------------------|----------------|
| (A) Leydig Cells | (I) Nutrition |
| (B) Hypothalamus | (II) FSH |
| (C) Sertoli cells | (III) Androgen |
| (D) Anterior pituitary | (IV) GnRH |

Choose the *correct* answer from the options given below:

1. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.11 Match List-I with List-II

List-I

Enzymes

(A) RNA polymerase I

(B) RNA polymerase III

(C) RNA polymerase II

(D) DNA polymerase

List-II

Products

(I) hnRNA

(II) Polymerization of deoxynucleotides

(III) rRNAs

(IV) tRNA, 5srRNA and snRNAs

Choose the *correct* answer from the options given below:

- 1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- 2. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- 3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
- 4. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.12 Which of the following enzymes join the sticky ends in recombinant DNA technology?

- 1. Exonuclease
- 2. Endonuclease
- 3. Ligase
- 4. Chitinase

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.13 Identify the incorrect statement about the genetic modification.

- 1. Genetic modification has helped to increase the post harvest losses.
- 2. Genetic modification has made crops more tolerant to abiotic stresses.
- 3. Genetic modification has increased efficiency of mineral usage by plants.
- 4. Genetic modification has reduced reliance on chemical pesticides.

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.14 Which one of the following is not true for a molecule that can act as a genetic material?

- 1. It should be able to generate its replica.
- 2. It should be chemically stable.
- 3. It should be catalytic and easily degradable.
- 4. It should be able to express itself in the form of Mendelian characters.

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.15 Which one of the following is a false fruit?

- 1. Mango
- 2. Papaya
- 3. Watermelon
- 4. Strawberry

Options 1. 1

2. 2

3. 3

4. 4

Q.16 Arrange the following in the order of their occurrence.

(A) Spermatids are transformed into spermatozoa

(B) Spermatogonia are multiplied by mitotic division

(C) Sperms are released from seminiferous tubules by the process of spermiation

(D) Sperm heads are embedded in sertoli cells

Choose the *correct* answer from the options given below:

1. (A), (B), (C), (D)

2. (B), (A), (D), (C)

3. (A), (D), (C), (B)

4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.17 Which one of the given species has more numbers than the combined total number of rest of the other given species?

1. Fungi

2. Algae

3. Lichens

4. Ferns and allies

Options 1. 1

2. 2

3. 3

4. 4

Q.18 Arrange the geological periods in order of occurrence from the recent to oldest one.

(A) Tertiary

(B) Permian

(C) Carboniferous

(D) Jurassic

Choose the *correct* answer from the options given below:

1. (A), (D), (B), (C)

2. (A), (C), (B), (D)

3. (B), (A), (D), (C)

4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.19 The sequencing of fragments using automated DNA sequencers in HGP, worked on the principle of the method that was developed by:

1. Charles Darwin

2. S.L. Miller

3. Erwin Chargaff

4. Frederick Sanger

Options 1. 1

2. 2

3. 3

4. 4

Q.20 Which one of the following is a result of convergent evolution?

1. The eye of octopus and eye of mammals

2. Vertebrate hearts

3. Fore limbs of cheetah and man

4. Vertebrate brains

Options 1. 1

2. 2

3. 3

4. 4

Q.21 The rate of biomass production is called productivity. It is expressed in terms of -

1. $\text{gm}^{-2} \text{yr}^{-1}$ or $(\text{kcal m}^{-2}) \text{yr}^{-1}$
2. $\text{gm}^{-2} \text{yr}^{-2}$ or $(\text{kcal m}^{-1}) \text{yr}^{-1}$
3. $\text{gm}^{-1} \text{yr}^{-1}$ or $(\text{kcal m}^{-2}) \text{yr}^{-2}$
4. $\text{gm}^{-1} \text{yr}^{-2}$ or $(\text{kcal m}^{-1}) \text{yr}^{-2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.22 Which of the following correctly expresses the equation for species- area relationships? Where(S=Species richness, A=Area, Z=slope of line (regression coefficient), C=Y- intercept

1. $\log S = \log C/A \log Z$
2. $\log S = \log C + Z \log A$
3. $\log S = \log C - Z \log A$
4. $\log S = Z \log C - \log (A)^2$

Options 1. 1

2. 2

3. 3

4. 4

Q.23 Apomixis is the -

1. Form of sexual reproduction that mimics sexual reproduction
2. Form of asexual reproduction that mimics asexual reproduction
3. Form of asexual reproduction that mimics sexual reproduction
4. Form of asexual reproduction that does not mimics sexual reproduction

Options 1. 1

2. 2

3. 3

4. 4

Q.24 Fruits develop without fertilization in few species, the process is called:

1. Parthenocarpy
2. Triple fusion
3. Dormancy
4. Syngamy

Options 1. 1

2. 2

3. 3

4. 4

Q.25 Which of the following are correct with reference to rDNA technology?

- (A) Exo-nuclease removes nucleotides from the ends of DNA
(B) Replication starts at ori
(C) High velocity microparticle of gold, coated with DNA, are used to transfer DNA in a method known as gene gun
(D) Endo-nuclease makes cut at specific positions within DNA
Choose the *correct* answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

Q.26 Match List-I with List-II**List-I****Organisms**

- (A) Barnacles *Balanus* and *Chathamalus* in the intertidal area
 (B) Orchid growing on a mango branch
 (C) Plants and herbivores
 (D) Mycorrhiza between fungi and roots of higher plants

List-II**Types of interaction**

- (I) Mutualism
 (II) Competition
 (III) Predation
 (IV) Commensalism

Choose the *correct* answer from the options given below:

- (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
- (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.27 Arrange the following hormones involved in correct sequence during menstrual cycle?

- (A) Estrogen
 (B) Progesterone
 (C) FSH
 (D) LH

Choose the *correct* answer from the options given below:

- (A), (B), (C), (D)
- (C), (A), (D), (B)
- (C), (D), (A), (B)
- (D), (B), (A), (C)

Options 1. 1

2. 2

3. 3

4. 4

Q.28 The eggs of the cuckoo (koel) have evolved to resemble crow's eggs in size and color. If the parasite bird has laid its eggs in the nest of host and lets the host to incubate them. This is an example of:

- Mutualism
- Competition
- Commensalism
- Brood parasitism

Options 1. 1

2. 2

3. 3

4. 4

Q.29 In an ecosystem, if a fish is feeding on zooplankton, it is a:

- Producer
- Primary consumer
- Secondary consumer
- Tertiary consumer

Options 1. 1

2. 2

3. 3

4. 4

Q.30 Match List-I with List-II

List-I

(A) Bt Cotton

(B) Method of Cellular defense

(C) Genetically engineered insulin (III) RNAi

(D) Gene therapy

List-II

(I) Functional ADA cDNA

(II) 'A' peptide

(IV) Insecticidal protein

Choose the *correct* answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.31 Which one of the followings is not a step in amplification of gene of interest using PCR?

1. Somatic hybridisation
2. Denaturation
3. Extension
4. Annealing

Options 1. 1

2. 2
3. 3
4. 4

Q.32 Match List-I with List-II

List-I

Evolutionary stages of man

(A) *Dryopethicus*(B) *Australopithecines*(C) *Homo habilis*(D) *Homo erectus***List-II**

Features

(I) Brain capacity 900 cc

(II) Hairy and walked like gorillas

(III) Brain capacity 650-800 cc

(IV) Lived in east African grasslands

Choose the *correct* answer from the options given below:

1. (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Where are IUDs placed in the human female reproductive sytem for contraception?

1. In either of the ovaries
2. In both the ovaries
3. In the uterus
4. In the vagina

Options 1. 1

2. 2
3. 3
4. 4

Q.34 If in a forest there were 570 sambhars last year and through reproduction 30 new sambhars were added, taking the current population to 600. Calculate the birth rate.

1. 5 offsprings per sambhar per year

2. 0.6 offsprings per sambhar per year
3. 0.052 offsprings per sambhar per year
4. 20 offsprings per sambhar per year

Options 1. 1

2. 2
3. 3
4. 4

Q.35 Which of the following are not associated to Lac Operon?

- (A) Lysozyme
- (B) β -galactosidase
- (C) Chitinase
- (D) Transacetylase

Choose the *correct* answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A) and (C) only
4. (B), and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.36 Arrange the following in order of their appearance, if a sperm starts travelling from testicular lobules?

- (A) Urethra
- (B) Vas deferens
- (C) Urethral meatus
- (D) Vasa efferentia

Choose the *correct* answer from the options given below:

1. (D), (B), (A), (C)
2. (A), (C), (D), (B)
3. (B), (A), (D), (C)
4. (D), (B), (C), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.37 Which of the following step is not the part of downstream processing?

1. Separation and purification
2. Product formulation with suitable preservatives
3. No quality control testing of product
4. Clinical trials of drugs

Options 1. 1

2. 2
3. 3
4. 4

Q.38 Arrange the following events in the order of their occurrence in tobacco plants to prevent infestation of nematode (*Meloidogyne incognitia*), based on the process of RNA interference.

- (A) Specific mRNA of the nematode is silenced
- (B) DNA produced sense and anti-sense RNA
- (C) Nematode specific genes were introduced into the host plant
- (D) dsRNA is formed and RNAi is initiated

Choose the *correct* answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3

Q.39 Which one of the followings is an example of ex situ conservation?

1. Seed bank
2. National park
3. Sacred groves
4. Wild life Sanctuary

Options 1. 1

2. 2
3. 3
4. 4

Q.40 The varieties of rice in India which are one of the richest in the world, they are approximately:

1. 2000 Varieties
2. 200 Varieties
3. 10,000 Varieties
4. 200,000 Varieties

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the following paragraph and answer the given questions.

Bio-fertilisers are organisms that enrich the nutrient quality of the soil like cyanobacteria, while there are organisms that help to control plant diseases and pests. The very familiar beetle, the ladybird and dragonflies are useful to get rid of aphids and mosquitoes, respectively. An example of microbial biocontrol agents that can be introduced to control butterfly caterpillars is certain bacteria, which are available in sachets in the form of dry spores. These spores are mixed with water and sprayed onto vulnerable plants. *Trichoderma* species that are very common in the root ecosystem, are very effective biocontrol agents of several plant pathogens. Baculoviruses attack insects and other arthropods. Usage of these viruses is desirable when an ecological sensitive area is being treated.

Which one of the following is used to control caterpillars of butterflies?

1. *Meloidogyne incognita*
2. *Agrobacterium tumefaciens*
3. *Bacillus thuringiensis*
4. *Haemophilus influenzae*

Options 1. 1

2. 2
3. 3
4. 4

Q.42 Read the following paragraph and answer the given questions.

Bio-fertilisers are organisms that enrich the nutrient quality of the soil like cyanobacteria, while there are organisms that help to control plant diseases and pests. The very familiar beetle, the ladybird and dragonflies are useful to get rid of aphids and mosquitoes, respectively. An example of microbial biocontrol agents that can be introduced to control butterfly caterpillars is certain bacteria, which are available in sachets in the form of dry spores. These spores are mixed with water and sprayed onto vulnerable plants. *Trichoderma* species that are very common in the root ecosystem, are very effective biocontrol agents of several plant pathogens. Baculoviruses attack insects and other arthropods. Usage of these viruses is desirable when an ecological sensitive area is being treated.

Trichoderma is a/an:

1. Free living fungi
2. Mycobiont element in a lichen.
3. Algal element in a mycorrhizae
4. Insect

Options 1. 1

2. 2
3. 3

Q.43 Read the following paragraph and answer the given questions.

Bio-fertilisers are organisms that enrich the nutrient quality of the soil like cyanobacteria, while there are organisms that help to control plant diseases and pests. The very familiar beetle, the ladybird and dragonflies are useful to get rid of aphids and mosquitoes, respectively. An example of microbial biocontrol agents that can be introduced to control butterfly caterpillars is certain bacteria, which are available in sachets in the form of dry spores. These spores are mixed with water and sprayed onto vulnerable plants. *Trichoderma* species that are very common in the root ecosystem, are very effective biocontrol agents of several plant pathogens. Baculoviruses attack insects and other arthropods. Usage of these viruses is desirable when an ecological sensitive area is being treated.

Which one of the following is not an advantage of *Nucleopolyhedrovirus* as biocontrol agents?

1. Narrow spectrum insecticidal applications
2. Species-specific viruses
3. Negative impacts on plants and birds
4. Helpful in conserving beneficial insects

Options 1. 1

2. 2
3. 3
4. 4

Q.44 Read the following paragraph and answer the given questions.

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A farmer wants to get rid of aphids, but does not want to use insecticide, Choose the best option for him?

1. Ladybird
2. Dragonflies
3. *Microsporum*
4. Statins

Options 1. 1

2. 2
3. 3
4. 4

Q.45 Read the following paragraph and answer the given questions.

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Find the odd one out of the given organisms?

1. Dragonflies
2. *Nostoc*
3. *Oscillatoria*
4. *Anabaena*

Options 1. 1

2. 2

3. 3

4. 4

Q.46 Read the following paragraph and answer the given questions.

Cancer detection is based on biopsy and histopathological studies. Techniques like radiography (use of X-rays), CT (computed tomography) and MRI (magnetic resonance imaging) are very useful to detect cancers of the internal organs. Do you know why mother's milk is considered very essential for the new-born infant? The yellowish fluid colostrum secreted by mother during the initial days of lactation has abundant antibodies (IgA) to protect the infant. Different species of *Plasmodium* (*P. vivax*, *P. malaria* and *P. falciparum*) are responsible for different types of malaria. Of these, malignant malaria caused by *Plasmodium falciparum* is the most serious one and can even be fatal. The exaggerated response of the immune system to certain antigens present in the environment is called allergy. The use of drugs like anti-histamine, adrenalin and steroids quickly reduce the symptoms of allergy. Drugs like barbiturates, amphetamines, benzodiazepines, and other similar drugs, that are normally used as medicines to help patients cope with mental illnesses like depression and insomnia, are often abused. Morphine is a very effective sedative and painkiller, and is very useful in patients who have undergone surgery.

Malignant malaria is caused by -

1. *Plasmodium malaria*
2. *Plasmodium vivax*
3. *Plasmodium falciparum*
4. *Plasmodium falciparum*

Options 1. 1

2. 2

3. 3

4. 4

Q.47 Read the following paragraph and answer the given questions.

Cancer detection is based on biopsy and histopathological studies. Techniques like radiography (use of X-rays), CT (computed tomography) and MRI (magnetic resonance imaging) are very useful to detect cancers of the internal organs. Do you know why mother's milk is considered very essential for the new-born infant? The yellowish fluid colostrum secreted by mother during the initial days of lactation has abundant antibodies (IgA) to protect the infant. Different species of *Plasmodium* (*P. vivax*, *P. malaria* and *P. falciparum*) are responsible for different types of malaria. Of these, malignant malaria caused by *Plasmodium falciparum* is the most serious one and can even be fatal. The exaggerated response of the immune system to certain antigens present in the environment is called allergy. The use of drugs like anti-histamine, adrenalin and steroids quickly reduce the symptoms of allergy. Drugs like barbiturates, amphetamines, benzodiazepines, and other similar drugs, that are normally used as medicines to help patients cope with mental illnesses like depression and insomnia, are often abused. Morphine is a very effective sedative and painkiller, and is very useful in patients who have undergone surgery.

A pregnant female is advised not to discard the yellowish fluid called colostrum, secreted during the initial days of lactation, because it is rich in:

1. IgA
2. IgE
3. IgG
4. IgM

Options 1. 1

2. 2

3. 3

4. 4

Q.48 Read the following paragraph and answer the given questions.

Cancer detection is based on biopsy and histopathological studies. Techniques like radiography (use of X-rays), CT (computed tomography) and MRI (magnetic resonance imaging) are very useful to detect cancers of the internal organs. Do you know why mother's milk is considered very essential for the new-born infant? The yellowish fluid colostrum secreted by mother during the initial days of lactation has abundant antibodies (IgA) to protect the infant. Different species of *Plasmodium* (*P. vivax*, *P. malaria*

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Which type of drug cannot quickly reduce the symptoms of allergy?

1. Anti-histamine
2. Adrenalin
3. Steroids
4. Benzodiazepines

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Read the following paragraph and answer the given questions.

Cancer detection is based on biopsy and histopathological studies. Techniques like radiography (use of X-rays), CT (computed tomography) and MRI (magnetic resonance imaging) are very useful to detect cancers of the internal organs. Do you know why mother's milk is considered very essential for the new-born infant? The yellowish fluid colostrum secreted by mother during the initial days of lactation has abundant antibodies (IgA) to protect the infant. Different species of *Plasmodium* (*P. vivax*, *P. malaria* and *P. falciparum*) are responsible for different types of malaria. Of these, malignant malaria caused by *Plasmodium falciparum* is the most serious one and can even be fatal. The exaggerated response of the immune system to certain antigens present in the environment is called allergy. The use of drugs like anti-histamine, adrenalin and steroids quickly reduce the symptoms of allergy. Drugs like barbiturates, amphetamines, benzodiazepines, and other similar drugs, that are normally used as medicines to help patients cope with mental illnesses like depression and insomnia, are often abused. Morphine is a very effective sedative and painkiller, and is very useful in patients who have undergone surgery.

Which one of the following is not used for detection of cancer?

1. Computed tomography
2. Biological response modifiers as α interferon
3. Magnetic resonance imaging
4. Biopsy

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the following paragraph and answer the given questions.

Cancer detection is based on biopsy and histopathological studies. Techniques like radiography (use of X-rays), CT (computed tomography) and MRI (magnetic resonance imaging) are very useful to detect cancers of the internal organs. Do you know why mother's milk is considered very essential for the new-born infant? The yellowish fluid colostrum secreted by mother during the initial days of lactation has abundant antibodies (IgA) to protect the infant. Different species of *Plasmodium* (*P. vivax*, *P. malaria* and *P. falciparum*) are responsible for different types of malaria. Of these, malignant malaria caused by *Plasmodium falciparum* is the most serious one and can even be fatal. The exaggerated response of the immune system to certain antigens present in the environment is called allergy. The use of drugs like anti-histamine, adrenalin and steroids quickly reduce the symptoms of allergy. Drugs like barbiturates, amphetamines, benzodiazepines, and other similar drugs, that are normally used as medicines to help patients cope with mental illnesses like depression and insomnia, are often abused. Morphine is a very effective sedative and painkiller, and is very useful in patients who have undergone surgery.

The drug, which is very effective sedative and painkiller and useful for patients who have undergone surgery is -

1. Morphine

- 2. Barbiturates
- 3. Benzodiazepines
- 4. Amphetamines

Options 1. 1

2. 2

3. 3

4. 4

